

Appendix A

Botanical Data

A.1 Shuna Botanical Data

The botanical remains in the following three tables was used to calculate the approximate contributions of the major components of the diet at Shuna. All three of the charts have been reformatted by myself, but are drawn completely from the work of Holden which is to be included in the forthcoming monograph on Shuna by Baird and Philip.

Table A.1: CARBONISED PLANT REMAINS FROM ESH SHUNA - Chalcolithic

Latin Name	Plant Part	Common Name	Count
Rumex cf. acetosella	nutlet	sheep's sorrel	1
Rumex sp.	nutlet	dock	1
cf. Rumex sp	nutlet	dock	1
cf. Polygonaceae	nutlet	knotweed family	2
cf. Chenopodium sp.	nutlet	goosefoot, etc.	1
Amaranthus sp.	seed	amaranth	1
Adonis sp.	achene	pheasants eye	10
Ranunculus sp.	achene	buttercup/crowfoot	1
Astragalus sp.	seed	vetch	2
cf. Astragalus sp.	seed	vetch	2
Vicia sp.	seed	vetch	1
Vicia/Lathyrus sp.	seed	vetch/tare/vetchling	13
Vicia/Lathyrus/Pisum sp.	seed	vetch/tare/vetchling	1
Trifolieae, Medicago type	seed	medlk	6
Trifolieae, Lotus/Trifolium type	seed	trefoil/clover	61
cf. Trifolieae, Lotus/Trifolium type	seed	trefoil/clover	2
Trifolieae, Trigonella type	seed		10
Scorpiurus muricatus L.	seed		290
Leguminosae indet.	seed	pea family	13
Leguminosae indet (large seeded)	seed	pea family	21
Leguminosae indet (small seeded)	seed	pea family	2
Malva sp.	seed	mallow	2
cf. Bupleurum sp.	fruit		1
Chrysanthemum sp.	achene		2

Continued on Next Page...

Table A.1 – Continued

Latin Name	Plant Part	Common Name	Count
cf. Thymelaea sp.	seed		2
Valerianella dentata (L.) Pollich	nutlet	corn salad	2
Compositae indet.	achene	daisy family	1
Lolium sp.	caryopsis	rye-grass	475
cf. Lolium sp.	caryopsis	rye-grass	12
Bromus sp.	caryopsis	brome	1
cf. Poa sp.	caryopsis	poa/meadow-grass	1
Phalaris type	caryopsis		19
Gramineae indet.	caryopsis	grass indet.	8
Gramineae (large grained)	caryopsis	large-grained grass	22
Gramineae (medium grained)	caryopsis	medium-grained grass	25
Gramineae (small grained)	caryopsis	small-grained grass	6
Cereal indet.	caryopsis	cereal indet.	11
Stipa sp.	caryopsis	stipa	3
cf. Stipa sp.	caryopsis	stipa	8
Scirpus maritimus L.	nutlet	sea club-rush	15
Scirpus cf. maritimus	nutlet	club rush	5
Scirpus sp.	nutlet	club-rushes	3
Scirpus setaceus L.	nutlet	bristle scirpus	192
Indeterminate	seed		12
Crops			
Ficus carica L.	seed	fig	17
cf. Ficus carica	seed	fig	8
Lathyrus sativus L.	seed	grass pea	20

Continued on Next Page...

Table A.1 – Continued

Latin Name	Plant Part	Common Name	Count
cf. <i>Lathyrus sativus</i>	seed	grass pea	7
<i>Vicia ervilia</i> (L.) Willd.	seed	vetch/tare	3
cf. <i>Vicia ervilia</i>	seed	vetch/tare	2
<i>Vicia faba</i> L.	seed	celtic bean	4
cf. <i>Vicia faba</i>	seed	celtic bean/horsebean	4
<i>Lens culinaris</i> Medicus	seed	lentil	125
cf. <i>Lens culinaris</i>	seed	lentil	3
<i>Linum usitatissimum</i> L.	seed	cultivated flax	1
<i>Olea europaea</i> L.	stone	olive	9
cf. <i>Pisum sativum</i>	seed	pea	1
<i>Triticum monococcum</i> L.	caryopsis	einkorn wheat	5
<i>Triticum monococcum/dicoccum</i>	caryopsis	einkorn/emmer	12
<i>Triticum monococcum/dicoccum</i>	glume base	einkorn/emmer	8
<i>Triticum monococcum/dicoccum</i>	single	einkorn/emmer	40
<i>Triticum monococcum/dicoccum</i>	spikelet fork	einkorn/emmer	2
<i>Triticum monococcum/dicoccum</i>	spikelet fork	einkorn wheat	1
<i>Triticum cf. monococcum</i>	caryopsis	emmer wheat	130
<i>Triticum dicoccum</i> Schrank	glume base	emmer wheat	1
<i>Triticum dicoccum</i> Schrank	caryopsis	emmer wheat	62
<i>Triticum dicoccum</i> Schubl.	glume base	emmer wheat	3
<i>Triticum dicoccum</i> Schubl.	caryopsis	emmer wheat	56
<i>Triticum cf. dicoccum</i>	glume base	emmer wheat	12
<i>Triticum cf. dicoccum</i>	spikelet fork	emmer wheat	3
<i>Triticum cf. aestivo-compactum</i>	caryopsis	bread/club wheat	1
<i>Triticum</i> sp.	caryopsis	wheat indet.	262

Continued on Next Page...

Table A.1 – Continued

Latin Name	Plant Part	Common Name	Count
Triticum sp.	glume base	wheat indet.	59
Triticum sp.	spikelet fork	wheat indet.	1
cf. Triticum sp.	caryopsis	wheat indet.	7
Triticum sp. (cf. free threshing)	caryopsis	free threshing wheat	1
Hordeum sativum indet.	caryopsis	barley indet.	104
cf. Hordeum sativum indet.	caryopsis	barley indet.	6
Hordeum sativum (hulled)	caryopsis	hulled barley	128
Hordeum sativum L. (hulled - straight)	caryopsis	hulled barley	82
Hordeum sativum L. (hulled - twisted)	caryopsis	hulled barley	48
Hordeum sp.	caryopsis	wild/cultivated barley	12
cf. Hordeum sp.	caryopsis	barley sp.	2
Cereal indet.	caryopsis	cereal indet.	5

Table A.2: CARBONISED PLANT REMAINS FROM ESH SHUNA - Early EB I

Latin Name	Plant Part	Common Name	Count
cf. Polygonaceae	nutlet	knotweed family	3
Amaranthus sp.	seed	amaranth	6
cf. Caryophyllaceae	seed		33
Adonis sp.	achene	pheasants eye	1
Fumaria sp.	fruit	fumitory	1
Astragalus sp.	seed	vetch	208
cf. Astragalus sp.	seed	vetch	3
Vicia/Lathyrus/Pisum sp.	seed	vetch/tare/vetchling/pea	1
Trifolieae, Melilotus type	seed	melilot	4
Trifolieae, Medicago type	seed	medik	18
Trifolieae, Lotus/Trifolium type	seed	trefoil/clover	128
Scorpiurus muricatus L.	seed		391
Trifolieae, Trigonella type	seed		4
Leguminosae indet (small seeded)	seed	pea family	10
Leguminosae indet (large seeded)	seed	pea family	2
Malva sp.	seed	mallow	16
cf. Malva sp.	seed	mallow	5
Compositae indet.	achene	daisy family	11
Lolium sp.	caryopsis	rye-grass	16
cf. Lolium sp.	caryopsis	rye-grass	6
Phalaris type	caryopsis		5
Gramineae indet.	caryopsis	grass indet.	10
Gramineae (large grained)	caryopsis	large-grained grass	6
Gramineae (medium grained)	caryopsis	medium-grained grass	18

Continued on Next Page...

Table A.2 – Continued

Latin Name	Plant Part	Common Name	Count
Gramineae (small grained)	caryopsis	small-grained grass	14
cf. Eleocharis sp.	nutlet	spike-rush	3
cf. Scirpus sp.	nutlet	club-rush	1
Scirpus setaceus L.	nutlet	bristle scirpus	43
Indeterminate	seed		19
Crops			
Vicia ervilia (L.) Willd.	seed	bitter vetch	3
cf. Vicia ervilia	seed	bitter vetch	1
Vicia faba L.	seed	celtic bean	1
Lens culinaris Medicus	seed	lentil	185
cf. Lens culinaris	seed	lentil	3
cf. Pisum sativum	seed	pea	4
Leguminosae indet (large seeded)	seed	pea family	18
Vitis vinifera L.	seed	vine	46
cf. Vitis vinifera	seed	vine	2
Olea europaea L.	stone	olive	21
Triticum monococcum L.	glume base	einkorn wheat	2
Triticum monococcum L.	spikelet fork	einkorn wheat	1
Triticum monococcum/dicoccum	caryopsis	einkorn/emmer	3
Triticum monococcum/dicoccum	glume base	einkorn/emmer	55
Triticum monococcum/dicoccum	single	einkorn/emmer	5
Triticum monococcum/dicoccum	spikelet fork	einkorn/emmer	4
Triticum cf. monococcum	spikelet fork	einkorn wheat	1
Triticum dicoccum Schubl.	caryopsis	emmer wheat	23
Triticum dicoccum Schubl.	glume base	emmer wheat	1

Continued on Next Page...

Table A.2 – Continued

Latin Name	Plant Part	Common Name	Count
<i>Triticum dicoccum</i> Schubl.	spikelet fork	emmer wheat	1
<i>Triticum dicoccum</i> Schrank	caryopsis	emmer wheat	19
<i>Triticum cf. dicoccum</i>	caryopsis	emmer wheat	30
<i>Triticum cf. dicoccum</i>	glume base	emmer wheat	1
<i>Triticum sp.</i>	caryopsis	wheat indet.	117
<i>Triticum sp.</i>	glume base	wheat indet.	6
<i>Triticum sp.</i>	spikelet fork	wheat indet.	5
<i>Triticum sp.</i>	caryopsis	wheat indet.	1
<i>cf. Triticum sp.</i>	caryopsis	hulled barley	5
<i>Hordeum sativum</i> (hulled)	caryopsis	hulled barley	4
<i>Hordeum sativum</i> (hulled - straight)	caryopsis	hulled barley	2
<i>Hordeum sativum</i> (hulled - twisted)	caryopsis	wild/cultivated barley	25
<i>Hordeum sp.</i>	caryopsis	cereal indet.	14

Table A.3: CARBONISED PLANT REMAINS FROM ESH SHUNA - Late EB I

Latin Name	Plant Part	Common Name	Count
Amaranthus sp.	seed	amaranth	3
Portulacca sp.	seed		1
cf. Caryophyllaceae	seed		1
Astragalus sp.	seed	vetch	15
Vicia/Lathyrus sp.	seed	vetch/rare/vetchling	3
cf. Vicial/Lathyrus sp.	seed	vetch/rare/vetchling	1
Vicia/Lathyrus/Pisum sp.	seed	vetch/rare/vetchling/pea	3
Trifolieae, Medicago type	seed	medik	1
Trifolieae, Lotus/Trifolium type	seed	trefoil/clover	51
Scorpiurus muricatus L.	seed		240
cf. Scorpiurus muricatus	seed		13
Leguminosae indet (large seeded)	seed	pea family	4
Malva sp.	seed	mallow	8
cf. Malva sp.	seed	mallow	2
cf. Chrysanthum sp.	achene		1
Hyoscyamus niger L.	seed	henbane	1
Plantago sp.	seed	plantain/ribwort	1
Lolium sp.	caryopsis	rye-grass	65
cf. Lolium sp.	caryopsis	rye-grass	16
Avena sp.	awn fragment	oat	1
Gramineae indet.	caryopsis	grass indet.	7
Gramineae (large grained)	caryopsis	large-grained grass	7
Gramineae (medium grained)	caryopsis	medium-grained grass	32
Gramineae (small grained)	caryopsis	small-grained grass	13

Continued on Next Page...

Table A.3 – Continued

Latin Name	Plant Part	Common Name	Count
<i>Scirpus maritimus</i> L.	nutlet	Sea club-rush	1
<i>Scirpus setaceus</i> L.	nutlet	bristle scirpus	19
Indeterminate	seed		8
Crops			
cf. <i>Ficus carica</i>	seed	fig	2
<i>Lens culinaris</i> Medicus	seed	lentil	16
<i>Vitis vinifera</i> L.	seed	vine	57
<i>Olea europaea</i> L.	stone	olive	24
<i>Triticum monococcum/dicoccum</i>	caryopsis	einkorn/emmer	15
<i>Triticum monococcum/dicoccum</i>	glume base	einkorn/emmer	7
<i>Triticum monococcum/dicoccum</i>	single	einkorn/emmer	7
<i>Triticum monococcum/dicoccum</i>	spikelet fork	eikorn/emmer	1
<i>Triticum cf. monococcum</i>	glume base	einkorn wheat	5
<i>Triticum cf. monococcum</i>	spikelet fork	einkorn wheat	8
<i>Triticum cf. monococcum</i>	caryopsis	emmer wheat	32
<i>Triticum dicoccum</i> Schubl.	caryopsis	emmer wheat	9
<i>Triticum cf. dicoccum</i>	glume base	emmer wheat	4
<i>Triticum cf. dicoccum</i>	spikelet fork	emmer wheat	5
<i>Triticum sp.</i>	caryopsis	wheat indet.	59
<i>Triticum sp.</i>	glume base	wheat indet.	46
<i>Triticum sp.</i>	spikelet fork	wheat indet.	13
<i>Triticum sp.</i>	tail grain	wheat indet.	1
cf. <i>Triticum sp.</i>	caryopsis	wheat indet.	5
<i>Hordeum sativum</i> indet.	caryopsis	barley indet.	4

Continued on Next Page...

Table A.3 – Continued

Latin Name	Plant Part	Common Name	Count
Hordeum sativum (hulled)	caryopsis	hulled barley	24
Hordeum sativum L. (hulled - straight)	caryopsis	hulled barley	2
Hordeum sp.	caryopsis	wild/cultivated barley	27
cf. Hordeum sp.	caryopsis	barley sp.	1

Appendix B

Faunal Data

B.1 Shuna Faunal Data

The faunal remains in the following three tables was used to calculate the approximate contributions of the major components of the diet at Shuna. All three of the charts have been reformatted by myself, but are drawn completely from the work of Croft which is to be included in the forthcoming monograph on Shuna by Baird and Philip.

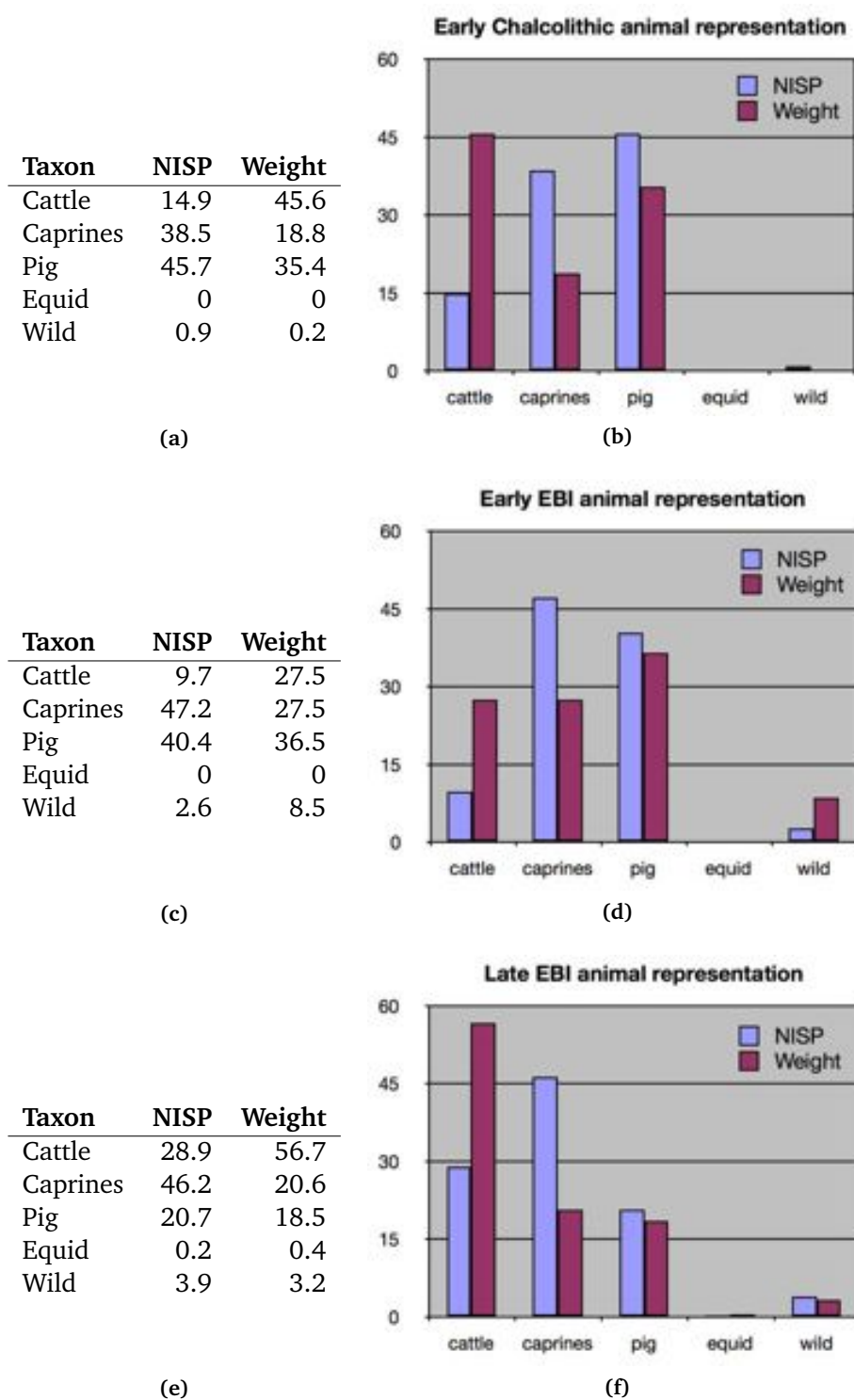


Figure B.1: Multi-period Animal Representation Data (Holden)

- (a) remains during Chalcolithic

(c) remains during EEBI

(e) remains during LEBI

(b) chart for Chalcolithic

(d) chart for EEBI

(f) chart for LEBI

Appendix C

Model Inputs

C.1 Variables

Model	
Site Details	
Name	Name of the settlement
Period	Period being examined
Population	Estimated population of the settlement
MCalsDay	Average daily MCals required per person
MeatPortion	The portion of the diet satisfied by meat
DomMeat	Percentage of the Meat Portion of the diet satisfied by meat from domestic sources
DairyUtil	The percentage of the potential dairy that will be utilised
DairyBool	Yes/No to a limit being placed on the percent of the dairy contribution in relation to the overall diet
DairyLimit	If dairy use is being limited, this is the limit (as a % of overall diet)

Table C.1: The model input variables required for the modelling process.

Animals	
Physiological Characteristics	
Name	Name of the animal being described
GestFeed	Feed Energy required while gestating
LactFeed	Feed Energy required while lactating
MaintFeed	Feed Energy required for maintenance
JuvFeed	Feed Energy per kg for juveniles (offspring)
Land	Food Value (MCals) of grazing land per Ha
Grain	Grams of grain (fed as supplement) per day
Fodder	Grams of straw and chaff per day
Days	Number of days animals fed grain and fodder
MeatVal	Food value of animal's meat (MCals per kg)
Usable	% of live weight that is usable as food
Kil	Average kill weight (kg)
WeanKg	Weight at weaning (kg)
AdultKg	Weight of average adult (kg)
Grow	Age (in days) of offspring
Death	Death Rate of offspring (%)
Coh	Conception rate (%)
SexMat	Age of sexual maturity (Months)
Breed	Number of years females can be bred
Offspring	Number of offspring per birthing event
Gest	Days of gestation
Dest	Days in-oestrous cycle
Lact	Days capable of lactating
Felis	Number of females servicable by one male
MilkGrams	Grams of milk produced per day
MilkMCals	Food Value of 1 kg of milk (MCals)
Percent	Percentage of the domestic meat portion of the diet that is to be satisfied by this animal

Table C.2: The animal input variables required for the modelling process.

Crops	
Physiological Characteristics	
Name	Name of the crop being described
Yield	Expected average yield of the crop (kg/Ha)
MCals	Food value of 1 kg of the final product
FallowRatio	The ratio of sown land to fallow land
FallowMCals	The food value of fallow land for grazing
FodderKg	The kg of straw and chaff produced per Ha
FodderMCals	The food value to animals of the fodder
Percent	Percentage of the domestic plant portion of the diet that is to be satisfied by this crop

Table C.3: The crop input variables required for the modelling process.

